Evaluation of Antiradical Activity of Different Cocoa and Chocolate Products: Relation with Lipid and Protein Composition

Chocolate antioxidant properties are often claimed; however, they are frequently different from the parent natural sources due to the industry or artisan transformation. In particular, antioxidant property of chocolate and cocoa are not adequately taken into consideration by consumers who normally make use of this food just for its flavor and taste properties. In this study, we have investigated the antioxidant capacity and total phenolic content of cocoa nibs, cocoa masses, and corresponding chocolate bars with different percentages of cocoa from different origins. The antioxidant capacity of the different samples was measured by two different assays [1,1-diphenyl-2-picryl-hydrazyl radical (DPPH) and ferric reducing antioxidant of potency (FRAP) tests]. The Folin–Ciocalteu reagent was used to assess the total phenolic content. The masses showed a higher antioxidant power than the nibs, and this has been attributed to the fact that in the nibs is still present the lipid part, which will form the cocoa butter. The influence of milk, whey, and soy proteins was also investigated. Our results showed that the extra dark cocoa bar, 100% cocoa chocolate, is the best in terms of total polyphenol content and in terms of antioxidant capacity according to the DPPH and FRAP tests. In addition, the bars of organic dark chocolate 80%, dark Tanzania 80%, and Trinidad 80% products are well performing in all respects. As highlighted by us, the antiradical properties of cocoa products are higher than many antioxidant supplements in tablets.